North Spark

A personalized app where kids on the spectrum create interest-based learning modules for cognitive, emotional, and social skills

For CSIS 3375 UX Design
in Web & Mobile App,
Section 001

A Requirements Document Submitted to
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by

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I. Requirements Gathering

For the NorthSpark project, we are conducting a requirements-gathering user study to better understand the real needs of autistic youth, their families, and the professionals who support them. Because this is a niche group with complex, context-dependent challenges, we chose semi-structured interviews to capture rich stories about daily routines, stress points, and existing support tools rather than relying on high-level survey data. The insights from these interviews will help us define both functional requirements (what the app should do) and non-functional requirements (how it should feel, e.g., sensory-friendly, predictable, and safe) so that NorthSpark fits naturally into users' lives. The following sections describe the objectives of the study, our interview sample and method, the interview guide, and how we analyzed the data and translated it into design requirements for the app

Purpose and Objectives for Requirements Gathering User Study

This section explains why we are conducting a requirements-gathering user study for NorthSpark and what we hope to learn from our participants. It outlines the core objectives that will guide our questions and help translate user needs into concrete app requirements.

- Understand the daily context of autistic youth, families, and clinicians so NorthSpark supports real-life routines (school, therapy, home, community) rather than abstract use cases.
- Identify key stressful or challenging moments where support tools could reduce overload, improve emotional regulation, or smooth transitions.
- Discover what tools, apps, or supports participants already use, what works well, and where current solutions fail or feel inadequate.
- Explore user goals and definitions of success (e.g., more independence, less stress, better communication) to align NorthSpark features with real outcomes that matter.
- Elicit preferences and constraints around safety, comfort, and accessibility (sensory load, language, visuals, privacy) to inform non-functional requirements.
- Capture differences between user roles (autistic youth, caregivers, clinicians, support workers) so that NorthSpark can support multiple perspectives where appropriate.

Choice of User Study and User Sample

This section describes the research method we chose for requirements gathering and why it is appropriate for our project. It also explains who our target participants are and how their perspectives will contribute to shaping NorthSpark.

Column	Content
Type of user study	Semi-structured interviews
Rationale for interviews instead of surveys	• Target users (autistic youth, their families, and clinicians) form a relatively niche group where depth

Column	Content
	and nuance are more important than a large sample size. • Semi-structured interviews let us follow up on specific experiences, sensory issues, or contextual factors that a fixed survey might miss.
Target user types (dual/multi-perspective)	• Autistic teens and young adults (primary users or direct beneficiaries of support). • Parents and caregivers of autistic youth. • Therapists, clinicians, or support workers involved in autism support (e.g., behavior consultants, counsellors, educational assistants).

Table 1:
Overview of User Study Design

This constitutes a multi-perspective approach: the same interview guide is adapted across different user roles to understand overlapping and distinct needs for NorthSpark.

User Study Interview

This section presents the interview guide we are using during our semi-structured interviews. It shows each question and briefly explains how the responses will help inform specific requirements for the app.

#	Interview question	How this supports NorthSpark
1	Can you describe a typical day related to autism support (school, therapy, home, community) and the moments that feel most stressful or challenging?	Reveals real contexts, peak-stress moments, and routines so we can derive requirements that fit daily life (e.g., when and where NorthSpark should be used).
2	What kinds of tools, apps, or supports are you currently using to help with organization, emotional regulation, or communication, and how well do they work for you?	Identifies existing solutions and gaps, informing which features are necessary, which are redundant, and where NorthSpark can provide unique value.
3	Thinking about those stressful moments, what would you most like a tool like NorthSpark to help with or make easier?	Connects pain points to desired outcomes and candidate features, helping prioritize requirements based on user-perceived impact.
4	What features or qualities would make a support app feel comfortable and safe to use (for example, sensory aspects, language, privacy, or visuals)?	Informs non-functional requirements such as sensory friendliness, clarity of language, predictability, and privacy settings.
5	From your perspective, what would success look like after using NorthSpark for a few months?	Provides concrete outcome criteria (e.g., less stress, more independence, smoother transitions) that can guide both design decisions and later evaluation metrics.

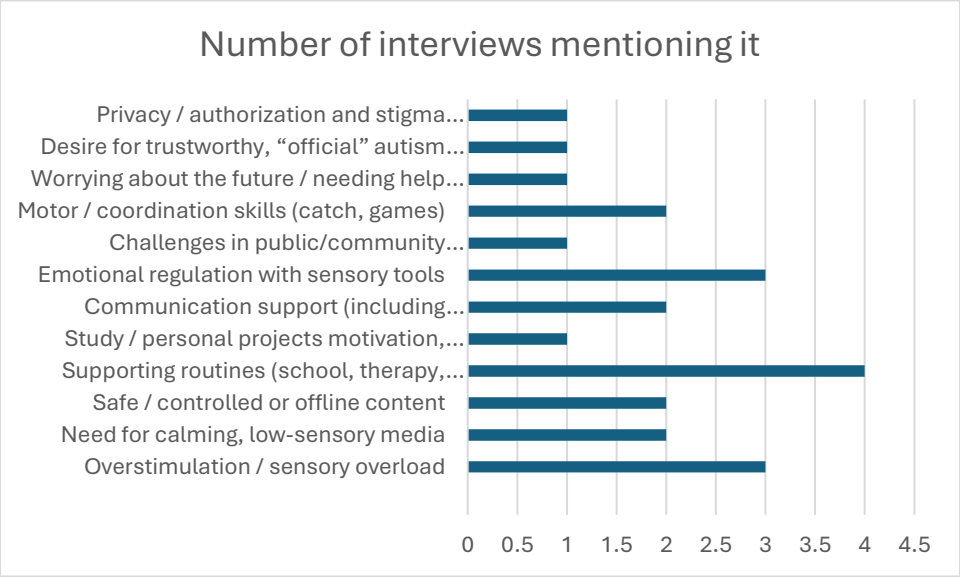
Table 2:
Interview Questions

Data Visualization and Analysis

In this section, we summarize how we organized and analyzed the interview data to extract requirements for NorthSpark. We combined qualitative thematic analysis with simple counts of how often key themes appeared across participants.

We conducted seven semi-structured interviews: six with parents/caregivers of autistic children and one with an autistic young adult (Spencer). We coded their responses into themes such as "overstimulation and sensory load," "screen time and content control," "offline or safe content," "supporting routines and study/study motivation," "skills development (communication, motor skills, social skills)," "future worries and life ownership," "getting help when needed," and "trust, privacy, and authorization." For a light quantitative view, we counted how many participants mentioned each theme and summarized this in a simple frequency table rather than a full statistical analysis, since the sample size is small.

Figure 1
Interview Results Graph



Theme	Number of interviews mentioning it	Interview #						
		1	2	3	4	5	6	7
Overstimulation / sensory overload	3	■		■	■			
Need for calming, low-sensory media	2	■			■			
Safe / controlled or offline content	2	■		■				
Supporting routines (school, therapy, sleep, study)	4		■	■	■	■		
Study / personal projects motivation, rewards	1						■	
Communication support (including non-verbal)	2				■	■		

Theme	Number of interviews mentioning it	Interview #						
		1	2	3	4	5	6	7
Emotional regulation with sensory tools	3	■			■	■		
Challenges in public/community understanding	1					■		
Motor / coordination skills (catch, games)	2		■		■			
Worrying about the future / needing help on demand	1						■	
Desire for trustworthy, "official" autism resources	1							■
Privacy / authorization and stigma concerns	1							■

Table 3
Interview Results

We attempted to recruit a diverse sample across our target user types. We contacted four therapists/clinicians; some declined and others could only schedule interviews during the class spring break (third week of March), so their input could not be included in this phase. We reached out to eight parents or caregivers, and six agreed to participate in interviews, providing rich perspectives on daily routines, calming strategies, and support needs at home, school, and in the community. In addition, we conducted one interview with an autistic young adult (Spencer), who is also a member of our design team; we chose to include his lived experience as formal data because first-person autistic perspectives are rarely represented in small student projects and add important insight into study pressure, future worries, and motivation.

Qualitatively, we examined each answer line by line and grouped similar ideas into themes such as "calming through music," "consistent daily schedule," "reward systems for sustained use," "making visual materials easier to use," "helping autistic youth step out of their comfort zone," and "getting advice from people who truly understand autism." We then wrote short theme summaries (1–2 sentences each) and linked them to possible app requirements—for example, a calming music feature, a simple routine planner with low-stim visuals, an in-app reward system, a trusted autism resources section, or clear privacy and authorization controls.

Insights from the Results

This section highlights the key findings from the interviews and explains how they shape the requirements for NorthSpark. Each point connects user needs to concrete implications for the app design.

- Overstimulation is a major challenge, especially in unstructured moments. Caregivers described situations where children run, climb, or cannot be left alone when overstimulated, which indicates that NorthSpark should include quick-access calming tools that can be used in-the-moment (e.g., one-tap calming activities or visuals).
- Calming, low sensory media is highly valued, but parents want control over content. One caregiver uses specific piano videos with falling lights on YouTube to soothe their child but worries about inappropriate recommendations, while another prefers simple songs with

minimal colors and stimuli, suggesting requirements for curated, low-stim, offline calming content inside the app.

- Parents want safe, trustworthy, and possibly offline experiences. Several responses emphasized wanting content that is "dedicated only to this type of calm content" and information from people who are qualified and can be trusted, which implies the app should have curated resources, clear sourcing, and options for offline use to avoid random online content.
- Daily structure and routines are important, but certain moments (like study time) create resistance. One caregiver described a detailed daily schedule (meals, school, ABA therapy, walks, screen time, bedtime), while another highlighted conflict around designated study time, suggesting that NorthSpark needs gentle routine-support features (visual schedules, timers, or transition supports) that are flexible and child-friendly.
- Success is defined as calm, engagement, and skill growth without total absorption. Caregivers see success when the child is calmer, still responsive when called, improving communication (verbal or nonverbal), or building motor skills like playing catch or eye-hand coordination, which means the app should focus on short, meaningful interactions rather than endless scrolling or passive watching.
- Physical tools (sensory toys, whiteboards, flash cards) are helpful but bulky. One caregiver uses dolls, swings, Mr. Potato Head, and a large whiteboard, and noted the inconvenience of physical flash cards, pointing to requirements for digital equivalents (e.g., customizable visual cards or activities) that are easy to bring anywhere.
- Design must balance "not overstimulating" with "interesting enough to get attention." Caregivers consistently asked for interfaces and activities that are visually calm yet still engaging, which directly informs non-functional requirements around color choices, animation, sound levels, and overall sensory design.
- Autistic youth also want motivation and streamlined support, not just "caregiver controls," highlighting the importance of user-facing features like rewards, reminders, and simple ways to ask for help.
- Some autistic youth express worries about the future and want help gaining "life ownership," as well as privacy and control over who can access their information, suggesting features for self-advocacy, trusted advice, and explicit authorization controls.

These insights will guide our next step, where we translate them into functional and non-functional requirements for NorthSpark, prioritizing calming tools, safe content, routine support, and sensory-friendly design.

II. Requirement Generation

Personas and Scenarios

This section introduces four personas that represent key user types for NorthSpark and shows short scenarios that highlight why they would use the app. These personas are grounded in our interview data and illustrate real motivations, contexts, and challenges related to autism support.

Persona	Scenario
"Maya", Parent of a Young Child Maya is a 35-year-old parent of a 6-year-old autistic boy who becomes easily overstimulated, especially in the late afternoon when he is tired. She currently uses YouTube to play specific calming piano videos with simple visuals but worries he might accidentally access inappropriate content.	**Calming Overstimulation at Home** When her child starts running around and climbing on furniture, Maya quickly opens NorthSpark to play a trusted calming routine: a low-stim piano video and simple breathing prompts. The content is offline and ad-free, so she does not have to monitor every click, and she can still get his attention when she calls his name.
"Sven's Dad", Routine-Focused Caregiver Sven's dad is in his early 40s and manages a very structured weekday schedule for his autistic son: meals, school, ABA therapy, walks, and screen time are all planned almost to the minute. He values consistency and exposure to real-world activities, but he worries about how his son will cope independently in the future	**Supporting Daily Routines and Independence** Before school, Sven's dad opens NorthSpark and quickly checks the visual schedule for the day, which includes icons for school, therapy, walk, and bedtime. Sven taps each step as he completes it, and over time he learns to follow the routine more independently with gentle reminders from the app instead of constant verbal prompts.
"Alex", Autistic Teen Who Likes Games Alex is a 15-year-old autistic teen who enjoys side-scrolling games and activities that improve eye–hand coordination. He finds some apps too noisy or visually overwhelming, but he is motivated by simple, skill-based games that help him practice coordination and focus.	**Practicing Skills in a Low-Stim Game** After school, Alex opens NorthSpark and chooses a side-scrolling coordination game with adjustable speed and muted colors. He plays short sessions that help him practice timing and motor skills without bright flashes or loud sounds, while the app quietly tracks his progress so he and his parents can see improvement over time.
"Lena", Caregiver Managing Study Time Lena is a caregiver of a 9-year-old autistic child who strongly resists designated "study time," often leading to conflict and stress for both of them. She uses sensory toys (doll, swing, Mr. Potato Head) and a big whiteboard, but carrying flash cards and physical materials everywhere is cumbersome.	**Making Study Time Easier and Less Stressful** During scheduled study time, Lena opens NorthSpark and selects a short, visual learning activity made from digital flash cards she created earlier. The app uses gentle animations and a calm color scheme to keep her child engaged without overload, and they can complete a few cards in a few minutes, making study time more manageable and repeatable throughout the day.

Table 4

Personas and Scenarios

Requirements Generation (from Personas and Scenarios)

Based on the analysis of these personas and scenarios, we generated the following requirements for NorthSpark. Each bullet is phrased as a "the system shall…" style requirement that can guide design and development.

Functional Requirements

- The system shall provide a library of calming audio-visual activities (e.g., piano music with simple visuals) that can be accessed quickly during moments of overstimulation.
- The system shall allow caregivers to download and play calming content offline, with no ads or external recommendations, to avoid exposing children to inappropriate material.
- The system shall support creation and editing of visual daily routines (e.g., school, therapy, walks, bedtime) using icons, simple text, and optional reminders.
- The system shall include short, low sensory games that target skills such as eye–hand coordination and basic motor control, with adjustable difficulty.
- The system shall let caregivers create and organize digital flash cards or learning materials (e.g., pictures, words, simple prompts) to replace bulky physical cards.
- The system shall track basic usage and skill progress over time (e.g., number of routines completed, game levels, practice sessions) and present this in a simple summary for caregivers and/or older users.

Non-Functional / UX Requirements

- The interface shall use calm, muted colors, limited motion, and minimal clutter to avoid overstimulation while still being engaging enough to hold attention for short sessions.
- Navigation shall be simple and predictable, with clear labels and consistent layouts so that users who do not learn new interfaces easily can still operate the app.
- The system shall provide clear privacy and safety safeguards, including caregiver control over what content is available and which features the child can access independently.

These requirements were derived by systematically reviewing each persona and scenario, identifying pain points, goals, and context, and translating them into functional and non-functional requirements for NorthSpark.

III. Synthesis of Requirements Gathering and Generation

This section combines insights from our user interviews and persona-based requirements generation into a single prioritized requirements list for NorthSpark. We group items into Must-haves (critical for an initial usable version) and Nice-to-haves (valuable but not essential for the first release).

Prioritized Requirements for NorthSpark

Must-have (critical) requirements

- Provide quick-access calming activities (e.g., simple piano videos, low-stim audio/visual routines) that caregivers can trigger during moments of overstimulation.
- Allow offline, ad-free access to curated calming and learning content so children are not exposed to random or inappropriate media.
- Support creation and use of visual daily routines (e.g., school, therapy, walks, study time, bedtime) with simple icons and predictable navigation.
- Include short, low sensory interactive activities or games that support skill development such as eye–hand coordination or simple motor skills.
- Enable caregivers to create and organize digital flash cards or similar visual supports to replace bulky physical materials.
- Use a sensory-friendly interface (calm colors, limited motion, clear layout) that is engaging but not overwhelming for autistic users.
- Provide basic caregiver control and safety features (e.g., locking certain areas, choosing available content, simple privacy settings).
- Provide simple, symbol- or text-based communication options that help children and autistic youth express basic needs, feelings, and "I need help" messages in real time, including during stressful moments at home, school, or in public.
- Keep core flows (starting a calming activity, checking a routine, asking for help) streamlined to a few clear steps so the app feels easy to use even when users are stressed, tired, or anxious about the future.

Nice-to-have (valuable but not essential for MVP)

- Track simple progress metrics over time (e.g., routines completed, game sessions, skill milestones) and show these in a caregiver summary.
- Offer optional templates or suggestions for calming routines and study sessions based on common scenarios reported in interviews (e.g., after-school transitions, bedtime wind-down).
- Allow lightweight personalization (favorite activities, preferred colors/sound levels) to better match individual sensory preferences.
- Provide access to trusted information or links from qualified professionals about autism, routines, and sensory strategies, curated inside the app.

- Offer an optional, low-pressure reward system (e.g., gentle badges or streaks) to give users extra incentive to keep using helpful routines or activities without turning the app into a gamified distraction.

Conclusions from the Requirements Process

Across both the interviews and persona analysis, we consistently saw that parents and caregivers need tools that are fast to access in stressful moments, safe and controlled in terms of content, and gentle enough not to overstimulate their child. At the same time, users define success not just as "keeping the child busy," but as calmer behavior, preserved responsiveness, and gradual improvement in communication and motor skills, so NorthSpark must prioritize short, purposeful interactions over endless entertainment.

The combination of data-driven insights and persona-based scenarios helped us distinguish a small set of Must-have features that form the minimum viable product (MVP), while clearly identifying additional Nice-to-have features that can be added in later iterations as time and resources allow. This process also confirmed that nonfunctional requirements such as sensory-friendly design, predictability, and safety are just as important as core functionality for our target users.

IV. AI Use Section

AI Tool Name	**Version Account Type**	**Specific Feature for which the AI Tool was Used**	**Value Addition** *(What value did you add over and above what AI did for you?)*
Perplexity	Education Pro	Asked AI to analyze competitor apps (TouchChat, CentralReach, Otsimo, Special Words, Choiceworks) and identify SEAT framework limitations vs North Spark's interest-driven approach	Provided specific app names and SEAT domains (Sensory, Escape, Attention, Tangible) from my research, plus clarified North Spark's dual-interface innovation that AI hadn't considered
Perplexity	Education Pro	Requested PACT framework analysis and value proposition drafting for autism learning app with flashcard database, AI recommendations, and parent/therapist dashboards	Gave detailed user personas (kids on spectrum + parents/therapists), activity context (personalized learning paths), and technology specs (AI image generation from special interests) to shape the output
Chatgpt	Free Account	Asked AI to surface themes form the 7 interviews we did	Provided the interview answers as input

AI Tool Name	Version Account Type	Specific Feature for which the AI Tool was Used	Value Addition *(What value did you add over and above what AI did for you?)*
Perplexity	Education Pro	Asked AI to refine the personas and scenarios we created	Provided the initial personas and scenarios created based on the interview results
Chatgpt	Free Account	Asked AI to refine and check for completeness of the requirements	Initialized Functional and Non Functional Requirements based on the personas and scenarios
Chatgpt	Free Account	Asked AI to further capture any other items that we missed	Initialized the must-haves and nice-to-haves based on the interviews and personas

Table 5

AI Usage & Value Addition

V. Work Log

Date	Number of Hours	Responsible Person	Description of Work Done
5 Jan 2026	0.5	All	Group Creation
12 Jan 2026	0.5	All	Brainstorming session for potential apps to be working on
15 Jan 2026	0.5	All	Initial app discussion with Priya
24 Jan 2026	0.5	All	Tasks discussion
28 Jan 2026	3	Dundee	Creation of project proposal – Intro and Background Research
29 Jan 2026	2.5	Dundee	Finalize project proposal – PACT, Value Proposition, Key Features GitHub repo creation
30 Jan 2026	1.5	Dundee	Finalize project proposal – Contract, AI Use and Worklog
31 Jan 2026	2	Dundee	Review and finalize initial draft
10 Feb 2026	1	All	Watch requirements gathering video
15 Feb 2026	1.5	All	Craft potential Interview questions for the user interview
2 Mar 2026	1	All	Narrow down questions to 5 ones for better focus
3 Mar 2026	1	All	Start listing down potential people to interview
4 Mar 2026	1	Dundee	Interview 2 people

Date	Number of Hours	Responsible Person	Description of Work Done
5 Mar 2026	1	Dundee	Do another interview for 2 people
6 Mar 2026	1	Spencer	Interview additional people
7 Mar 2026	2	Dundee	Do initial draft of the Requirement Document. Consolidate and analyze interview results. Update Requirements Generation section
8 Mar 2026	2.5	Dundee	Continue updating Requirements Document. Update Synthesis section
9 Mar 2026	3	All	Finalize Requirements Document. Incorporate interview 5, 6 & 7. Submit to document to blackboard

Table 6

Work Log

VI. References

Interaction Design Foundation. (2026). *UX storyboards: Ultimate guide*. https://www.interaction-design.org/literature/article/ux-storyboards

Maze. (2025, February 24). *UX research objectives: Defining research direction*. https://maze.co/blog/ux-research-objectives/

Nielsen, J. (2019, September 23). *Requirements gathering for better user experience*. UsabilityGeek. https://usabilitygeek.com/requirements-gathering-user-experience-pt1/

Smagorinsky, S. (2016, September 1). *How to design for autistic people*. Smashing Magazine. https://smart-interface-design-patterns.com/articles/design-autism/

Tiimo. (2022, February 1). *Sensory processing routines for autistic adults*. https://www.tiimoapp.com/resource-hub/sensory-processing-routines

VII. Appendix

List of AI Prompts

Prompts
Given our project to create a personalized app for kids on the spectrum to learn, we identified 3 major challenges such as low motivation to learn for non-preferred activities, escapism behaviours and not taking into account special interests and dislikes, find relevant literature that would support each of those challenges.
Find other competitor apps aside from what we know so far like TouchChat and CentralReach and compare offerings to North Spark features.
Refine our write-up in a way that does not omit the flow of our paragraphs as well as the main points we are conveying, check for grammatical errors and polish sentences that do not provide clarity.
For the key features, organize the flow of presenting each item in a way that does not convey overwhelming functions considering the important points about our app such as the SEAT framework and taking into account the special interests and dislikes.
Here are transcripts from seven interviews with autistic youth and caregivers about their daily routines, stressors, and current tools. Please read them and identify the main themes and patterns in their needs, challenges, and goals, summarized in clear bullet points.
Based on the following interview themes and draft personas/scenarios for our NorthSpark autism-support app, please refine each persona and its scenario. Make them more specific, realistic, and clearly linked to the user needs and motivations described in the interviews.
Using these refined personas and scenarios for the NorthSpark app, review the functional and non-functional requirements we drafted. Check for completeness, remove duplicates, and rewrite them as clear 'The system shall…' statements that trace back to the personas' needs.
Here are our final requirements for the NorthSpark app plus a summary of interview and persona insights. Please group them into two lists: Must-have (critical for an MVP) and Nice-to-have (valuable but not essential), and briefly justify why each requirement belongs in that group.

Table 7

List of AI Prompts